Just Wiser

Buy-vs-Rent Decision Support Tool

Problem, Research and Objective

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Live application: https://wiser-gray.vercel.app/
GitHub repository: https://github.com/mlftsh030/wiser

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1 Introduction

Buying a home is one of the largest financial decisions an individual or household can make. The decision is often presented as a simple comparison between a monthly mortgage payment and monthly rent. In reality, the decision involves several financial, personal and long-term considerations.

Just Wiser is a decision-support tool designed to help users reason through the buy-versus-rent decision using their own circumstances.

The purpose of the application is not to tell a user that buying or renting is universally better. Instead, it calculates the financial consequences of both options, considers different scenarios, and then provides an explanation of which option appears more suitable given the information supplied.

The principle behind the system is:

The software calculates the decision. AI helps explain the decision.

The detailed implementation of the financial model, decision engine, application architecture, AI integration, testing and deployment is covered separately in Document 2.

2 The Problem

The main problem is that the buy-versus-rent decision is more complicated than comparing two monthly payments.

A prospective buyer needs to consider:

- whether the household can comfortably afford the property;
- the size of the required upfront contribution;
- transaction costs associated with purchasing;
- the monthly cost of owning the property;
- the length of time the household expects to remain in the property;
- the potential growth in property prices;
- future rental increases;
- the opportunity cost of money invested into the property;
- the financial safety provided by remaining savings; and
- personal preferences such as flexibility, stability and ownership.

This creates a problem for users who may have enough income to qualify for a property but do not necessarily have enough financial resilience to make the purchase a sensible decision.

Similarly, renting may appear cheaper in the short term while producing a different long-term financial outcome.

Just Wiser therefore treats the problem as a **decision-support problem** rather than a simple affordability calculation.

3 South African Context

The application was designed specifically with the South African financial environment in mind.

Interest rates, inflation, property transaction costs and household expenses can materially affect the outcome of a housing decision. The South African Reserve Bank's monetary policy environment therefore provides important context for the calculations used by the application.

At the time of development, the South African Reserve Bank reported a policy rate of 7% and inflation of 5.0%. The Bank's inflation target remains 3%. These conditions demonstrate why assumptions about future interest rates, inflation and housing costs should not be treated as certain predictions.

Property purchases can also involve transfer duty and other transaction costs. SARS provides a progressive transfer-duty structure, including a zero-duty band for lower-value properties and increasing rates at higher property values.

These factors motivated the inclusion of transaction costs and scenario analysis rather than relying only on the advertised property price.

4 Research

The research for Just Wiser focused on understanding what should actually influence a buy-versus-rent decision.

The research considered South African economic conditions, housing-related costs, taxation and the broader financial trade-offs associated with property ownership.

The main sources consulted included the South African Reserve Bank, South African Revenue Service and Statistics South Africa.

The research did not attempt to predict the future. Instead, it was used to identify the financial variables and risks that a useful decision-support tool should expose to the user.

5 Key Findings

The research produced several important findings.

5.1 Affordability is not enough

Being able to make a monthly mortgage payment does not necessarily mean that purchasing is financially safe.

A buyer may have insufficient savings remaining after the deposit and transaction costs, leaving little protection against unexpected expenses.

For this reason, Just Wiser considers both monthly affordability and the remaining emergency buffer.

5.2 The purchase price is not the full cost

The cost of owning a property extends beyond the mortgage.

The model considers costs such as:

- mortgage repayments;
- municipal rates;
- insurance;
- maintenance; and
- levies where applicable.

The initial purchase can also involve transaction costs such as transfer duty and conveyancing-related expenses.

5.3 Time horizon matters

Buying and selling property involves significant transaction costs. As a result, a short expected period of ownership can produce a very different result from a long-term ownership period.

Just Wiser therefore asks users how long they expect to remain in the property and uses this information when comparing the two strategies.

5.4 Renting has an opportunity cost

Renting does not simply mean spending money instead of building an asset.

If a renter has capital that could have been invested, or if the renter saves the difference between renting and owning, those investments can produce a future value.

The comparison therefore considers the potential investment value available to a renter.

5.5 Long-term predictions are uncertain

Future property growth, rental growth and investment returns cannot be known with certainty.

Rather than presenting one future outcome as fact, Just Wiser uses multiple scenarios to show how the decision could change under different assumptions.

6 Objective

The objective of Just Wiser is to create an understandable decision-support tool that combines personal financial information with housing and long-term financial assumptions to help users reason about whether buying or renting is more appropriate for their circumstances.

The system should:

1. collect relevant financial and personal information;
2. calculate the financial implications of buying;
3. calculate the financial implications of renting;
4. compare the two strategies over an appropriate time horizon;
5. test how the result changes under different scenarios;

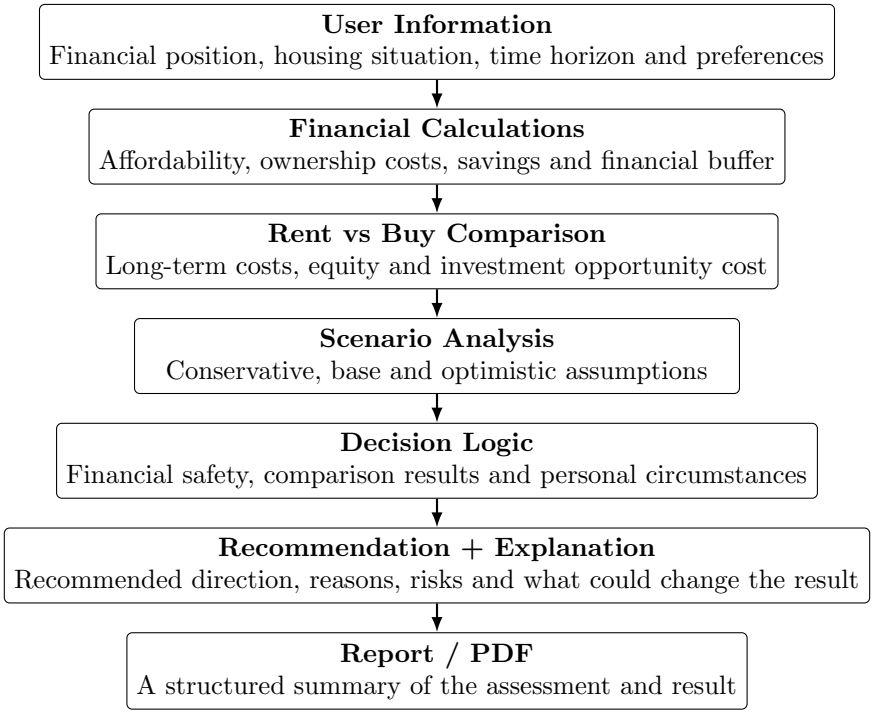
6. identify financial risks and constraints;
7. incorporate relevant personal circumstances and preferences; and
8. communicate the result in a way that a non-technical user can understand.

The goal is therefore not simply to produce a numerical answer, but to help the user understand **why** the result was reached and **what could change it**.

7 Proposed Solution

Just Wiser uses a structured process that moves from the user's information to financial calculations, comparison, scenario analysis and finally a recommendation.

The overall process can be summarised as follows:



The process separates calculation from interpretation. Financial values are calculated deterministically before a recommendation is produced.

This is important because the recommendation should be traceable to the user's inputs and the assumptions used by the model.

8 What Just Wiser Considers

The assessment collects information across four broad areas.

Area	Examples of information considered
Financial position	Monthly household income, essential expenses, existing debt repayments and accessible savings.
Housing	Current rent, target property price, upfront contribution, interest rate and loan term.
Personal circumstances	Expected length of stay and likelihood of moving within the next few years.
Preferences and risk	Importance of flexibility, importance of owning a home, comfort with financial uncertainty and personal priorities.

This allows the tool to consider the user's situation rather than treating every household as financially identical.

9 Decision Philosophy

Just Wiser does not use a simple weighted score where positive preferences can cancel out serious financial risks.

For example, a strong preference for home ownership should not be able to compensate for a household being unable to meet the projected monthly ownership cost.

The decision process therefore follows a hierarchy:

1. **Financial safety:** determine whether buying is financially viable at all.
2. **Financial comparison:** compare the long-term financial position of buying and renting.
3. **Personal circumstances:** use factors such as flexibility and expected length of stay to refine the result.

This means that personal preferences can influence a financially reasonable decision, but they cannot override major affordability or financial-safety problems.

The system can produce outcomes such as:

- RENT FOR NOW
- BUYING MAY WORK
- BUYING LOOKS STRONGER

These are decision-support conclusions, not financial advice.

10 Limitations

Just Wiser is intentionally a decision-support tool rather than a financial advisory system.

Its results are subject to several limitations:

- future interest rates, property growth, rental growth and investment returns are uncertain;
- transaction and professional costs are approximated;

- the model does not attempt to represent every possible South African property transaction;
- regional differences in property markets and rental markets are not modelled in detail;
- the mortgage model assumes a relatively standard home loan;
- advanced products such as access bonds and complex repayment strategies are not modelled;
- the model does not replace a bank's credit assessment;
- user inputs are self-reported and may be inaccurate; and
- the result should not be interpreted as regulated financial advice.

The tool is therefore best understood as a structured way of thinking about the decision rather than a prediction of what will happen.

11 Outcome

The result of the project is a working web application that takes a user's financial and personal information and produces a structured buy-versus-rent assessment.

The application provides:

- a guided questionnaire;
- financial calculations;
- rent-versus-buy comparisons;
- scenario analysis;
- a structured recommendation;
- explanations of the main factors affecting the result; and
- a downloadable report in PDF format.

The application is available online:

https://wiser-gray.vercel.app/

The source code and implementation documentation are available on GitHub:

https://github.com/mlftsh030/wiser

Readers interested in the technical implementation can refer to **Document 2: Design, Implementation and Extended Features**.

12 Conclusion

The buy-versus-rent decision cannot be reduced to whether a household can afford a monthly mortgage payment.

A useful decision-support system must consider the user's financial safety, the full cost of ownership, the opportunity cost of renting, the expected period of ownership and the uncertainty surrounding future financial conditions.

Just Wiser was developed around this principle.

The application combines deterministic financial calculations, scenario analysis and structured decision logic to provide a recommendation that is explainable rather than simply producing a score.

Its purpose is ultimately to help users ask better questions about a major financial decision:

Can I afford this?
What am I giving up by choosing it?
How long do I expect to stay?
How sensitive is the decision to future conditions?
What would need to change for the other option to become better?

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